

Model Notes for Bragg-Interferometer Phase Noise Driven by Raman Velocity-Selection Noise

Project note for the local Python tool

1 Purpose

This document defines the first implementation of a calculation tool for the phase noise induced in a first-order Bragg Mach–Zehnder atom interferometer by shot-to-shot fluctuations of the velocity distribution selected by a Raman velocity-selection stage.

The present scope is intentionally simplified:

- atomic species: ^{87}Rb ,
- interferometer sequence: $\pi/2 - \pi - \pi/2$,
- Bragg order: first order,
- Bragg pulses: finite-duration Gaussian pulses,
- Raman-selected ensemble: one-dimensional Gaussian detuning distribution,
- shot noise source: Gaussian noise on the center detuning of the selected distribution,
- spatial cloud size and position–velocity correlations: neglected.

2 Experimental Default Configuration

The default values embedded in the user interface match the current experimental discussion:

Quantity	Default value
Atomic species	^{87}Rb
Bragg order	$n = 1$
Pulse sequence	$\pi/2 - \pi - \pi/2$
Bragg beam-splitter duration	25 μs
Bragg mirror duration	50 μs
Bragg pulse shape	Gaussian
Dark time	8 ms
Selected distribution FWHM	12 kHz
Reference shot-noise RMS	4 kHz
Nominal center detuning	0 kHz relative to Bragg resonance

The value “15 kHz for $+\hbar k$ in the Raman configuration” is treated as contextual calibration information, not as a parameter that directly enters the current Bragg detuning-domain solver. The tool uses the detuning relative to the nominal Bragg resonance of the selected class.

3 Two-Level Bragg Pulse Model

For a given Doppler-equivalent two-photon detuning δ and laser phase set to zero, the first-order Bragg pulse is approximated as an effective two-level system in the basis

$$\{ |0\rangle, |2\hbar k\rangle \}.$$

The Hamiltonian is

$$\frac{H(t)}{\hbar} = \frac{1}{2} \begin{pmatrix} -\delta & \Omega(t) \\ \Omega(t) & \delta \end{pmatrix}, \quad (1)$$

where $\Omega(t)$ is the real Gaussian two-photon Rabi frequency envelope.

The Gaussian envelope is written as

$$\Omega(t) = \Omega_0 \exp\left(-\frac{t^2}{2\tau^2}\right). \quad (2)$$

The code supports several conventions for the user-entered pulse duration. Internally, all durations are converted to the RMS envelope width τ .

The peak Rabi frequency is calibrated from the target pulse area,

$$\theta = \int_{-N_\sigma\tau}^{+N_\sigma\tau} \Omega(t) dt = \Omega_0\tau\sqrt{2\pi} \operatorname{erf}\left(\frac{N_\sigma}{\sqrt{2}}\right), \quad (3)$$

with $\theta = \pi/2$ for the beam splitter and $\theta = \pi$ for the mirror.

4 Free Evolution

In the same rotating frame, the dark-time evolution between pulses is represented by

$$U_{\text{free}}(T) = \begin{pmatrix} e^{+i\delta T/2} & 0 \\ 0 & e^{-i\delta T/2} \end{pmatrix}. \quad (4)$$

This term captures the detuning-dependent differential phase accumulated between the momentum states in the simplified two-level picture.

5 Ensemble Averaging

The Raman-selected atoms are represented by a Gaussian detuning distribution

$$g(\Delta; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta - \mu)^2}{2\sigma^2}\right], \quad (5)$$

where μ is the shot-dependent center detuning and

$$\sigma = \frac{\text{FWHM}}{2\sqrt{2\ln 2}}. \quad (6)$$

The shot-to-shot Raman-selection noise is modeled as a fluctuation of the center,

$$\mu_{\text{shot}} \sim \mathcal{N}(\mu_0, \sigma_{\text{noise}}^2). \quad (7)$$

In the implementation, ensemble averages over $g(\Delta; \mu, \sigma)$ are evaluated by Gauss–Hermite quadrature.

6 Fringe Coefficient and Total Interferometer Phase

Let

$$U_{\text{BS}}, \quad U_{\text{M}} \quad (8)$$

denote the finite-duration beam-splitter and mirror pulse unitaries at phase zero for a fixed detuning class Δ .

For the full sequence with free evolution, the state just before the last pulse is

$$|\psi_{\text{pre}}\rangle = U_{\text{free}}(T) U_{\text{M}} U_{\text{free}}(T) U_{\text{BS}} |0\rangle. \quad (9)$$

If the phase of the final beam splitter is scanned by ϕ , the population in the $|0\rangle$ output port can always be written as

$$P_0(\phi) = P_{\text{off}} + 2 \operatorname{Re} \left[c e^{i\phi} \right]. \quad (10)$$

The complex fringe coefficient c is computed explicitly by the code. The fringe phase is then defined by

$$\phi_{\text{tot}} = -\arg(c). \quad (11)$$

For an ensemble, the averaged coefficient is

$$\bar{c}(\mu) = \int g(\Delta; \mu, \sigma) c(\Delta) d\Delta, \quad (12)$$

and the corresponding total interferometer phase is

$$\Phi_{\text{tot}}(\mu) = -\arg[\bar{c}(\mu)]. \quad (13)$$

7 Operational Definition of the Bragg Diffraction Phase

The current tool defines a *pulse-only diffraction phase* operationally by suppressing dark-time phase accumulation while keeping the same finite-duration Bragg pulses:

$$|\psi_{\text{pre}}^{(\text{diff})}\rangle = U_{\text{M}} U_{\text{BS}} |0\rangle. \quad (14)$$

The same fringe-coefficient extraction is then applied to obtain

$$\Phi_{\text{diff}}(\mu) = -\arg[\bar{c}_{\text{diff}}(\mu)]. \quad (15)$$

This is a model-dependent but well-defined quantity: it isolates the phase offset generated by finite-duration off-resonant Bragg pulses within the present solver. It should not be confused with a higher-order multi-state diffraction phase model unless extra momentum states are included explicitly.

8 Phase-Noise Outputs

The tool reports two complementary mappings from Raman center-noise RMS to phase noise:

1. a linearized propagation,

$$\sigma_{\phi} \approx \left| \frac{d\Phi}{d\mu} \right| \sigma_{\text{noise}}, \quad (16)$$

evaluated at the nominal center μ_0 ,

2. a Monte Carlo estimate obtained by drawing many realizations of μ_{shot} and computing the RMS phase excursion.

Both estimates are produced for

$$\Phi_{\text{tot}} \quad \text{and} \quad \Phi_{\text{diff}}. \quad (17)$$

9 Important Assumptions and Limits

The first implementation is rigorous within its chosen model, but it does *not* yet include:

- coupling to higher Bragg momentum orders,
- spatially varying intensity or wavefront effects,
- atomic position distribution,
- mean-field interactions,
- explicit Raman pulse dynamics or the physical origin of the shot noise,
- correlations between the center fluctuation and the selected width.

Accordingly, the present tool should be interpreted as a controlled first-order calculation of how shot-to-shot fluctuations of the selected velocity class propagate into Bragg-sequence phase offsets and phase noise.

10 Files

The Python interface is organized around:

- `main.py`: launches the desktop application,
- `bragg_phase_noise/config.py`: user-facing model configuration,
- `bragg_phase_noise/physics.py`: pulse propagation and interpolators,
- `bragg_phase_noise/analysis.py`: ensemble averaging and phase-noise calculations,
- `bragg_phase_noise/ui.py`: Tkinter + Matplotlib interface.